

Une approche par graphes des tests conformes en grande dimension utilisant des e-values basées sur la centralité, avec application à la détection d'anomalies

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Abstract

La détection d'anomalies peut être formulée naturellement comme un test d'hypothèse statistique : étant donné un ensemble de données d'entraînement échantillonnées à partir d'une distribution inconnue P , on souhaite déterminer si une nouvelle observation est issue de P ou d'une distribution $Q \neq P$. Ce problème est notoirement difficile, car spécifier un modèle paramétrique sous l'hypothèse nulle est souvent irréaliste dans les contextes modernes de grande dimension. L'inférence conforme offre un cadre puissant pour ce type de test sans hypothèse de distribution, en fournissant des p -valeurs valides sous des hypothèses minimales Bates et al. (2023). Dans ce travail, nous proposons un nouveau test conforme basé sur une représentation en graphe des k plus proches voisins (k-nn) des données d'entraînement. Notre méthode exploite des mesures de centralité dans le graphe pour construire une e -value quantifiant la conformité d'un point candidat par rapport à l'ensemble d'apprentissage. Cette e -value est ensuite transformée en une p -valeur conforme, ce qui permet d'obtenir un test valide à taille finie pour détecter des changements de distribution et des anomalies. L'approche par graphe k -NN est particulièrement adaptée aux données de grande dimension, car elle permet de capturer les structures de variété intrinsèquement de faible dimension et offre une manière flexible de définir des scores de conformité reflétant la géométrie locale des données. Nous évaluons notre méthode sur des jeux de données synthétiques et réels, y compris des données textuelles de grande dimension, en démontrant sa capacité à détecter des changements de distribution subtils tout en contrôlant rigoureusement l'erreur de type I. Les principaux avantages de notre approche sont les suivants : (i) elle ne nécessite pas de modèle statistique prédéfini pour tester l'hypothèse nulle, ce qui est crucial pour les applications en intelligence artificielle, (ii) elle est mathématiquement rigoureuse, et (iii) elle est simple à implémenter en pratique avec seulement quelques lignes de code.

Keywords: Test d'Hypothèses; e-values; Détection d'Anomalies.

1 Introduction

Anomaly detection is a fundamental problem in statistics and machine learning, with applications ranging from quality control and network intrusion detection to monitoring of scientific experiments and language model outputs. At its core, anomaly detection can be framed as a statistical hypothesis test: given a collection of training data points $X_1, \dots, X_n \sim P$, we wish to determine whether a new observation X_{n+1} comes from the same distribution P or from a different distribution $Q \neq P$. This corresponds to testing the null hypothesis

$$H_0 : X_{n+1} \sim P \quad \text{vs.} \quad H_1 : X_{n+1} \sim Q, \quad Q \neq P.$$

However, specifying P explicitly is rarely possible in modern settings, where data can be high-dimensional, heavy-tailed, and structured in complex ways. Classical parametric tests are thus inapplicable, and even nonparametric tests may fail without strong modeling assumptions.

The theory of *e-values* Ramdas and Wang (2024); Vovk et al. (2025) provides a principled, distribution-free approach to this problem. Combining the theory of *e-values* with the ideas underpinning the new field of Conformal Prediction and Conformal *p-values* Angelopoulos and Bates (2021); Bates et al. (2023), *e-values* allow to construct non-conformity scores that easily yield statistically valid finite-sample testing procedures under the sole assumption of exchangeability. Exact type-I error control going along the construction of these tests, makes them ideal for out-of-distribution (OOD) detection and other tasks where false positives must be rigorously controlled.

In this paper, we introduce a novel approach to conformal anomaly detection that leverages a k -nearest-neighbor (k-nn) graph representation of the data. We view the training set as a weighted graph, where nodes are data points and edges encode proximity in feature space. Using this representation, we define a new class of *e-values* Ramdas and Wang (2024); Vovk et al. (2021); Wang and Ramdas (2022); Vovk et al. (2025) based on graph-theoretic notions of centrality, such as closeness and betweenness. Intuitively, points that lie near the "center" of the data manifold should be assigned higher conformity scores, while points associated with abnormally small centrality are more likely to be anomalous. Our *e-value* is then converted into a conformal *p-value*, yielding a valid test of H_0 that adapts to the intrinsic geometry of the data.

The k-nn graph viewpoint has several key advantages in high dimensions. First, it provides a natural way to approximate the underlying data manifold, allowing the test to exploit low-dimensional structure without explicitly estimating densities. Second, graph centrality captures global information about the data distribution, beyond local density estimates. This makes our method particularly sensitive to subtle distributional shifts that affect global geometry rather than just local neighborhoods.

We validate our approach through experiments on both synthetic and real-world datasets, including high-dimensional textual data. Our results show that the proposed method achieves competitive or superior power compared to state-of-the-art conformal and nonconformal anomaly detection methods, while maintaining exact finite-sample validity. Beyond its practical performance, our work contributes a general framework for constructing model-free statistical tests from graph-based representations, opening new perspectives for robust, distribution-free inference in modern machine learning.

2 Background

2.1 Previous work on anomaly detection and testing

Classical anomaly detection. Early approaches treat anomalies as points in low-density regions. Distance- and density-based methods include k NN distance scoring, Local Outlier Factor (LOF) Breunig et al. (2000), one-class SVM Schölkopf et al. (2001), and Isolation Forest Liu et al. (2008). These techniques remain competitive, but their guarantees typically depend on smoothness or margin assumptions and degrade in high dimensions without additional structure (e.g., manifold assumptions).

Deep and representation-based approaches. Neural methods learn representations where anomalies are separable: autoencoder reconstruction scores, self-supervised pretext tasks, and energy-based or likelihood-ratio surrogates. While powerful in practice, pure likelihood from deep generative models can be miscalibrated for OOD Nalisnick et al. (2019). Subsequent work uses contrastive representations or density-ratio estimation to improve robustness, but distributional guarantees remain limited without resampling or calibration layers.

Two-sample testing without explicit models. A large literature develops nonparametric tests for $H_0 : P = Q$: kernel MMD Gretton et al. (2012), energy distance Székely and Rizzo (2005),

classifier two-sample tests (C2ST) Lopez-Paz and Oquab (2017), and Stein-discrepancy tests (KSD) Liu et al. (2016). These tests enjoy asymptotic validity and can be powerful in high dimensions when the chosen kernel or critic captures the salient structure. However, finite-sample calibration and selective inference in adaptive pipelines typically require resampling (permutation/wild bootstrap) or additional correction.

Conformal p -values for outlier detection. Conformal inference yields finite-sample *distribution-free* p -values under exchangeability by calibrating a nonconformity (or conformity) score on held-out data. For outlier detection, conformal p -values provide valid level- α tests without specifying P Bates et al. (2023). Variants include inductive and cross-conformal schemes, Mondrian conformal for group-wise calibration, and conformalization of classical scores (e.g., k NN, LOF, density surrogates), enabling exact type-I error control with power driven by the choice of score. Conformal methods have also been developed for time series and sequential monitoring via block-exchangeability or online recalibration Fedorova et al. (2012); Vovk et al. (2015).

e -values. An e -value is a nonnegative statistic with unit-or-less expectation under H_0 , enabling simple level- α tests via the threshold $1/\alpha$ and powerful *anytime-valid* and sequential combinations Vovk et al. (2021). Connections between conformal p -values and e -values allow mutual conversion and hybrid designs. In multiple testing, e -values support FDR control through procedures such as e-BH under independence or certain dependence structures, complementing classical BH Benjamini and Hochberg (1995); Wang and Ramdas (2022); Ignatiadis et al. (2023). This line of work is attractive for adaptive pipelines because e -values compose and average naturally across data splits or modalities.

Graph and manifold perspectives. Graph-based nonconformity scores leverage k NN graphs to approximate low-dimensional manifolds embedded in high dimensions, capturing local density and global geometry via centralities, commute times, or diffusion distances Perozzi et al. (2014); Coifman and Lafon (2006). Calibrating such scores conformally leads to tests whose validity is distribution-free while power adapts to intrinsic dimensionality. Our work contributes to this direction by defining an e -value from graph centrality and converting it into a conformal p -value, thereby combining finite-sample validity with geometry-aware power.

Multiple testing and selective inference. In large-scale screening, BH controls FDR under independence or PRDS dependence Benjamini and Hochberg (1995); Benjamini and Yekutieli (2001). Conformal p -values and e -values can be integrated with BH/e-BH to deliver error control in pipelines that learn representations or scores on one split and calibrate on another, mitigating selection bias. Recent work explores knockoff-style and sample-splitting strategies for feature selection and OOD detection with distribution-free guarantees; our approach follows the split-calibration paradigm while focusing on graph-derived scores.

2.2 Conformal p -values and e -values: a brief introduction

Statistical hypothesis testing classically relies on the concept of a p -value: a random variable $p(X)$ whose distribution under the null hypothesis H_0 is stochastically larger than the uniform distribution on $[0, 1]$. This ensures that the test that rejects when $p(X) \leq \alpha$ has type-I error at most α . However, constructing valid p -values typically requires strong distributional assumptions (e.g., Gaussianity) or access to the likelihood ratio between H_0 and H_1 .

Conformal p -values. Conformal inference Bates et al. (2023); Jin and Candès (2023) provides a distribution-free way to build valid p -values under the sole assumption that the data are exchangeable. The key idea is to compute a *conformity score* $S(x)$ for each data point, measuring how "typical" x is

relative to the training data. For a candidate point x_{new} , one compares its score $S(x_{\text{new}})$ to the scores of a calibration set, yielding the *conformal p-value*

$$p(x_{\text{new}}) = \frac{1 + \#\{i : S(X_i) \leq S(x_{\text{new}})\}}{1 + n_{\text{cal}}}.$$

By symmetry, these p -values are valid for any underlying distribution P , guaranteeing exact finite-sample control of type-I error. This makes conformal methods highly attractive for tasks such as anomaly detection and out-of-distribution (OOD) testing, where the null distribution is unknown.

e -values. An alternative way to quantify evidence is through an e -value Vovk et al. (2021); Wang and Ramdas (2022); Ramdas and Wang (2024). Formally, an e -value is a nonnegative random variable E satisfying $\mathbb{E}_{H_0}[E] \leq 1$. Intuitively, E plays the role of a "betting score": values much larger than 1 indicate evidence against H_0 . Testing with e -values is simple: reject H_0 whenever $E \geq 1/\alpha$, which guarantees a level- α test by Markov's inequality.

An advantage of e -values is their flexibility: they can be combined and updated sequentially, allowing for online or anytime-valid inference. Moreover, a valid p -value can be obtained from an e -value via the transformation

$$p = \min\left(1, \frac{1}{E}\right),$$

ensuring a direct connection between the two notions. In our work, we exploit this property: we design a new e -value based on graph centrality, and then convert it into a conformal p -value to obtain a finite-sample valid test.

3 Rigorous empirical testing using k-nn-graphs and centrality-based e -values

e -values can be interpreted as quantitative measures of the strength of evidence against the null hypothesis: the larger their value, the stronger the evidence in favor of the alternative hypothesis. Concretely, the e -value corresponds to the realization of the e -variable once the data are observed.

Definition 1. An e -variable E for $\mathcal{P}$ is a random variable taking values in $[0, \infty]$ that satisfies $\mathbb{E}^{\mathbb{P}}[E] \leq 1$ for all $\mathbb{P} \in \mathcal{P}$. An e -variable E is exact if $\mathbb{E}^{\mathbb{P}}[E] = 1$ for all $\mathbb{P} \in \mathcal{P}$.

We now formally introduce the notion of an e -variable, using a practical example: anomaly detection. In this setting, we assume we only have access to a calibration sample generated under the law $\mathbb{P}$ corresponding to the null hypothesis H_0 — for instance, a dataset consisting exclusively of healthy individuals in a medical context. The goal is to design a nonparametric testing procedure that can assess whether a new observation (or a new sample) can be considered consistent with the reference distribution or, on the contrary, should be flagged as potentially coming from an alternative distribution that signals an anomaly.

3.1 Theoretical underpinnings

Let the calibration data be $X_1, X_2, \dots, X_n$ and the test sample $X_{n+1}, X_{n+2}, \dots, X_{n+m}$. Under exchangeability of the combined calibration and test data, we construct nonnegative test statistics $L(X_1), L(X_2), \dots, L(X_{n+m})$ that preserve this exchangeability property. Let us define $L(X_i) = 1/C(X_i)$, $i = 1, 2, \dots, n$, where $C(X_i)$ denotes the centrality of X_i with respect to the calibration data. The larger L , the more likely the alternative hypothesis is. We then obtain the e -value for test data point j as

$$e_j = \frac{(n+1)L(x_{n+j})}{\sum_{i=1}^n L(x_i) + L(x_{n+j})} \quad (1)$$

We have the following result regarding the e -values defined before.

Proposition 1. *Let us define*

$$L(X_i) = \frac{1}{C(X_i)}, \quad i = 1, 2, \dots, n,$$

where $C(X_i)$ denotes the centrality of X_i with respect to the calibration data. Then, the e -values as defined in Equation (1) is an e -variable under $\mathbb{P}$ of H_0 .

The proof is deferred to the Appendix.

3.2 Multiple testing with e -values

In the previous section, we have shown that the expression of the e -variable given by equation (1) satisfies the expectation property under the null hypothesis, which makes it admissible for use in the e -BH (e-Benjamini–Hochberg) procedure. The algorithm 1, detailing this procedure, is presented below. This procedure is inspired by the theory of conformal p -values presented in Bates et al. (2023).

Algorithm 1 Multiple testing selection with e -values (e-BH procedure)

Input: Calibration data $\{X_i\}_{i=1}^n$, test data $\{X_{n+j}\}_{j=1}^m$ and a level $\alpha \in (0, 1)$.

1. Compute the test statistics $L(X_i)$ for $i = 1, \dots, n$ and $L(X_{n+j})$ for $j = 1, \dots, m$.
2. If necessary, compute scores R from the statistics L (R may be equal to L if L is nonnegative).
3. Generate $e_j = \frac{(n+1)R(x_{n+j})}{\sum_{i=1}^n R(x_i) + R(x_{n+j})}$ for $j = 1, \dots, m$.
4. Sort e_j from largest to smallest and obtain $e_{(k)}$ for $k = 1, \dots, m$.
5. Find the largest k^* such that $e_{(k)} \geq \frac{m}{\alpha k}$.

Output: Rejection set of H_j , $\mathcal{R} = \{j : e_j \geq \frac{m}{\alpha k^*}\}$.

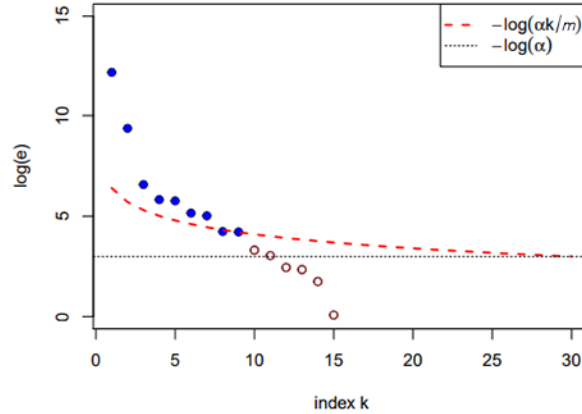


Figure 1: The ordered e -values and the (red) line above which they must lie for a test to be rejected by the e-Benjamini–Hochberg procedure.

4 Numerical experiments

This section presents the results obtained across all experiments conducted. The cases studied are grouped by category in order to highlight recurring behaviours as well as the limitations of our methodology. The associated figures serve as support for the analysis.

4.1 Experimental Results

The following experiments illustrate a range of settings, from clear successes to partial failures, that together delineate the conditions under which the proposed method is effective. In all cases, the calibration set consists of 1,000 samples drawn from the normal population, and the test set contains 50 anomalous instances mixed with unseen normal samples. All experiments use `all-mpnet-base-v2` embeddings unless otherwise stated.

4.1.1 Roleplay.

We consider a chat channel dedicated to a Medieval Fantasy universe, where out-of-roleplay intrusions (e.g., modern technical vocabulary) must be flagged. The method achieves a **100% detection rate**, with anomaly scores averaging 33.77 against near-zero scores for normal messages. The thematic gap between medieval and technical registers produces a clean geometric separation in the embedding space.

4.1.2 Phishing.

We evaluate detection of fraudulent corporate emails (financial scams, credential requests) embedded in a stream of legitimate professional communications. This setting is particularly challenging: the average intra-cluster distance within the normal population is negligible (≈ 0.014). Under this configuration, the method achieves a **100% detection rate** with anomaly scores exceeding 1,000, successfully distinguishing, for instance, “*Urgent wire transfer*” from the semantically adjacent but benign “*Urgent meeting*”.

4.1.3 Code Obfuscation.

We address detection of malicious or obfuscated Python scripts concealed within a corpus of clean source code. Clean code exhibits strong structural regularity (average intra-cluster distance of 0.008). The method achieves a **100% detection rate**, flagging even syntactically subtle patterns such as `__import__('os')`, which deviate only marginally from the structural norm of the calibration set. This confirms that `mpnet` embeddings encode sufficient structural information to distinguish obfuscated from idiomatic code.

4.1.4 Cheat Detection.

We consider detection of aim-assisted players in a multiplayer shooter, based solely on behavioural statistics (Kill/Death ratio, accuracy, score per minute, headshot count), without access to the cheat software itself. Features are normalised via `StandardScaler` prior to graph construction, and conformity scores are derived from a k-nn graph. The method achieves an **86% detection rate** (43/50), with flagged players obtaining scores exceeding 1,500 against a detection threshold of approximately 20. The 14% false negative rate is attributable to *subtle cheaters* who activate the software only intermittently, producing statistics indistinguishable from those of skilled legitimate players.

4.1.5 Medical Triage.

We evaluate the detection of life-threatening patient queries (e.g., “*Crushing chest pain*”, “*Facial paralysis*”) within a stream of routine consultation requests. This setting introduces a qualitatively different challenge: the boundary between normal and anomalous is not geometric but semantic, with benign and critical conditions sharing identical anatomical vocabulary (e.g., “*Sore throat*” vs. “*Swelling throat*”). We obtain a **76% detection rate** (38/50). Clearly distinct emergencies (e.g., “*Haemorrhage*”, “*Cardiac arrest*”) are reliably isolated (scores > 80), while keyword-overlapping cases

remain difficult to separate, suggesting that supervised augmentation would be necessary in a clinical deployment.

4.1.6 Summary.

Table 1 consolidates detection performance across all five experiments. Results confirm that the method is most effective when anomalies induce a clear geometric discontinuity in the embedding space, and degrades gracefully as the semantic overlap between normal and anomalous populations increases.

Table 1: Detection performance across all experimental settings. n_{cal} : calibration set size; n_{anom} : number of anomalies in the test set.

Task	n_{cal}	n_{anom}	Detection Rate
Roleplay	1,000	50	100%
Phishing	1,000	50	100%
Code Obfuscation	1,000	50	100%
Cheat Detection	1,000	50	86%
Medical Triage	1,000	50	76%

4.2 An Instructive Failure: Online Toxicity

We report an experiment in which the method produces systematically inverted results, as it provides insight into a fundamental limitation of the geometric approach.

4.2.1 Setting.

The task consists of detecting toxic messages (e.g., “Noob”, “Report him”) within a video game chat stream. The calibration set comprises 1,000 gameplay-related messages (e.g., “Rush point A”, “Buy smoke”, “Need healing”), and the test set contains 50 generic toxic messages (e.g., “Loser”, “You suck”, “Get lost”).

4.2.2 Result and Analysis.

The algorithm correctly identified the messages most distant from the geometric centre of the calibration set — yet flagged **100% of polite players** as anomalous, the exact inverse of the intended behaviour. This failure is not a numerical error but a *distributional misspecification*: in the specific game considered, toxic language had become the statistical norm, rendering polite messages the geometric outliers.

This case illustrates a fundamental limitation of any unsupervised, geometry-based anomaly detector: the method detects *distributional deviance*, not *semantic harm*. When the undesirable behaviour constitutes the majority of the data, the null hypothesis is misspecified by construction. Addressing such settings would require either (i) a supervised or semi-supervised formulation that encodes a notion of harm independently of frequency, or (ii) an external prior on what constitutes normative behaviour, which falls outside the distribution-free framework considered here.

4.3 Discussion

The results presented in the previous section show consistent and reproducible behaviours of the *e-values* framework when applied to high-dimensional data. This section analyses these results, distinguishing situations favourable to the *e-values* approach from those that reveal its limitations.

First, as can easily be noticed, successful experiments share a common characteristic: anomalies correspond to structural or semantic deviations that manifest as low overlap in the representation space. In these cases, the geometric modeling of normality allows the construction of a graph in which normal observations occupy dense regions, while anomalies appear at the periphery or in weakly connected regions.

These results confirm that when an anomaly manifests as a geometric discontinuity in the representation space, *e-values* constitute a viable choice for unsupervised detection, while preserving the statistical guarantees established by the original study.

The results obtained on medical data illustrate an intermediate scenario in which the normal population forms extremely dense clusters. In this context, some anomalies are embedded within the global structure, leading to *e-values* that mislead the model.

This behaviour shows that when normality is highly concentrated, the available space for distinguishing deviations becomes sufficiently small that the boundary between normal values and anomalies becomes blurred.

The toxicity case highlights a fundamental limitation of the studied approach. Toxic messages do not constitute a geometric anomaly: they are often central in the representation space. The approach fails to isolate them, producing low *e-values*.

This observation suggests that conformal *e-values* are only well-suited when the anomaly is correlated with an identifiable geometric structure in the representation space. When the anomaly is primarily linked to social or contextual phenomena, supervised or semi-supervised approaches may be more appropriate.

5 Conclusion and Future Work

In this work, we proposed a new approach to high dimensional testing based on empirical data under the null hypotheses only. The originality of our approach stems from the combination of the conformal *e-values* framework Ramdas and Wang (2024) with a graphical approach using centrality, or a monotone function thereof, as a *p-value*. The main advantages of our approach are that (i) it does not need a prescribed statistical model for testing the null, which is key for applications in AI, (ii) it is mathematically rigorous and (iii) it is straightforward to implement in practice using a few lines of code only.

The numerical results show that the proposed approach works remarkably well when anomalies correspond to clearly identifiable structural or semantic deviations. In such situations, the graphical approach effectively isolates anomalous observations and produces *e-values* consistent with the assumptions of the theory. Conversely, these results also reveal the limitations of the approach. When the anomaly is not a geometric discontinuity, but is instead rooted in pragmatic or social context, unsupervised detection via *e-values* proves inadequate. This failure contributes to refining the conditions under which *e-values* are relevant or not.

Several directions can be envisaged to extend this work. One avenue involves investigating finer calibration methods for similarity parameters, which could potentially adapt to the local density of the data. Another direction concerns the use of hybrid approaches combining unsupervised *e-values* and supervised models, in order to handle non-geometric anomalies.

In summary, this work contributes to extending the scope of conformal *e-values* in unsupervised anomaly detection, while providing some insights into their potential and their limitations in, e.g. high-dimensional natural language processing tasks.

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337 A Proof of Proposition 1

338 **Proposition 1.** *Let us define $L(X_i) = \frac{1}{C(X_i)}$, $i = 1, 2, \dots, n$, where $C(X_i)$ denotes the centrality of*
 339 *X_i with respect to the calibration data. Then, the e-values as defined in Equation (1) is an e-variable*
 340 *under $\mathbb{P}$ of H_0 .*

Proof. . Let $X_{n+j} = X_0$, we have the e-variable

$$E_0 = \frac{(n+1)L(X_0)}{\sum_{i=0}^n L(X_i)}.$$

341 Since $L(X_i) = \frac{1}{C(X_i)}$ for $i = 0, 1, \dots, n$ are exchangeable, we have

$$\begin{aligned} \mathbb{E}^{\mathbb{P}}[L(X_0) \mid \sum_{i=0}^n L(X_i)] &= \mathbb{E}^{\mathbb{P}}[L(X_1) \mid \sum_{i=0}^n L(X_i)] \\ &= \mathbb{E}^{\mathbb{P}}[L(X_n) \mid \sum_{i=0}^n L(X_i)], \end{aligned}$$

342 which implies

$$(n+1)\mathbb{E}^{\mathbb{P}}[L(X_0) \mid \sum_{i=0}^n L(X_i)] = \sum_{i=0}^n L(X_i). \quad (2)$$

343 Using equation (2), we obtain

$$\begin{aligned} \mathbb{E}^{\mathbb{P}}[E_0] &= \mathbb{E}^{\mathbb{P}} \left[\frac{(n+1)L(X_0)}{\sum_{i=0}^n L(X_i)} \right] \\ &= \mathbb{E}^{\mathbb{P}} \left[\mathbb{E}^{\mathbb{P}} \left[\frac{(n+1)L(X_0)}{\sum_{i=0}^n L(X_i)} \mid \sum_{i=0}^n L(X_i) \right] \right] \\ &= \mathbb{E}^{\mathbb{P}} \left[\frac{(n+1)}{\sum_{i=0}^n L(X_i)} \mathbb{E}^{\mathbb{P}}[L(X_0) \mid \sum_{i=0}^n L(X_i)] \right] \\ &= 1. \end{aligned}$$

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□